Chenyuan Zhang

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Research Summary

I am a neurosymbolic AI researcher working on how autonomous agents make sequential decisions, and on how agents reason about the goals and beliefs of other agents. My work combines **symbolic planning** and **knowledge representation** with learned perception and **large language models**, so that long-horizon reasoning in uncertain, partially observable environments stays interpretable and verifiable. At Monash I led the reasoning and planning workstream of the **DARPA ANSR**-funded HARNESS programme, taking goal recognition from formal problem definition through to flight on physical UAVs. My background in psychology also lets me test agent models against **controlled human experiments**, not only benchmarks. My work appears at AAMAS, ICAPS, KR, EMNLP, RA-L and CogSci, and received the **AAMAS 2024 Pragnesh Jay Modi Best Paper Award**.

Research Directions

Agent Decision Making

How an agent plans and acts over long horizons in complex, partially observable environments. I build **state abstractions** on top of learned perception and plan hierarchically over them, so that autonomous agents can carry out long-horizon search and action in unknown environments: *NEUSIS* (RA-L 2025, **co-first author**) is a compositional perception–reasoning–planning framework for complex UAV search missions, deployed and validated on physical platforms. In parallel I characterise what **large language models** can and cannot contribute as planning components: *Mini-BEHAVIOR-Gran* (**EMNLP 2026**) shows a **U-shaped** effect of instruction granularity on language-guided embodied agents — instructions that are too coarse and too fine both hurt — and *What We Talk About When We Talk About LLM Planning* argues that "LLM planning ability" conflates two distinct abilities that should be evaluated separately. My doctoral work embedded classical planners inside **Bayesian** models of intent and tested them against **controlled human experiments** (CogSci 2023, 2025), which is why I can evaluate agent systems against real decision behaviour rather than benchmark scores alone.

Multi-Agent Reasoning

How an agent infers the goals and beliefs of others, and acts on that inference. I formalised **goal recognition as Bayesian inference** with a planner as the likelihood model, and showed that *timing* is a strong and previously overlooked signal (ICAPS 2023; AAMAS 2024, **Best Paper Award**). I then extended recognition to the settings that deployment demands: **active** recognition, where the observer acts to resolve ambiguity (KR 2025); **hierarchical** goals (*A Probabilistic Framework for Hierarchical Goal Recognition*, **KR 2026**); and **partial observability** (*Neurosymbolic Active Goal Recognition in Partially Observable Environments*, **AAMAS 2026**). Building on this, I work on how agents model *others* and decide accordingly: *Modeling Higher-order Human Beliefs Using the Justified Perspective Model* (**CHI 2025**) gives a symbolic account of higher-order belief ("I think that you think…"), a computable form of theory of mind for multi-agent settings, and *Proactive Assistance Agent with Online Goal Recognition* (**ICAPS 2026**) puts online recognition inside an assistive agent so that it intervenes before the person asks. On the ARC Discovery project I paired learned behaviour models with symbolic planning to predict diverse human behaviour under weak supervision; supervised Master's projects extend the thread to **multi-agent reinforcement learning** and imitation learning. Current work scales hierarchical reasoning to multi-agent environments.

Research Experience

June 2026 – **Research Fellow (Level B), Faculty of Information Technology, Monash University** in Neuro-Symbolic AI. *Vision & Language for Autonomous AI (VL4AI) Lab.*

- Retained on **internal Faculty funding** to continue the neurosymbolic programme after the DARPA project concluded, with scope to set my own research agenda.
- Scaling **hierarchical symbolic reasoning** to longer horizons and to **multi-agent** settings, and characterising what **large language models** can and cannot contribute as planning components in transparent decision-support systems.

June 2024 – **Honorary Research Fellow, University of Melbourne** in Artificial Intelligence. *AI and Autonomy Lab*, School of Computing and Information Systems, and *Complex Human Data Hub*, Melbourne School of Psychological Sciences.

- Supervise Master's research projects on neurosymbolic AI, **multi-agent reinforcement learning**, imitation learning, and symbolic models of higher-order belief, mentoring students from problem formulation through to publication; more than five papers to date.

Mar 2024 – June 2026 **Research Fellow (Level B), Monash University** in Neuro-Symbolic UAV Autonomy.

- Led the **symbolic reasoning and planning** workstream of **HARNESS: Hierarchical Abstractions and Reasoning for Neuro-Symbolic Systems**, funded by the **DARPA Assured Neuro-Symbolic Learning and Reasoning (ANSR)** program. Designed the **hierarchical abstraction and planning layer** above learned perception that lets UAVs search unknown environments, and co-first-authored the resulting **NEUSIS** architecture.
- Led the programme's **goal recognition** thread from formal problem definition through to flight on physical UAV platforms, extending recognition to the **active**, **hierarchical**, and **partially observable** settings that deployment demands.
- Integrated the system on **ROS 2** and validated it in **AirSim** before **sim-to-real** transfer; coordinated the perception and hardware sub-teams, and contributed the technical results and demonstrations behind a **successful project extension**.

June 2024 – June 2025 **Research Fellow (Level B), Monash University** in Human-Robot Interaction.

- Human-modelling researcher on the **ARC Discovery Project** *Human Models for Accelerated Robot Learning and Human-Robot Interaction*, pairing **learned behaviour models** with **symbolic planning** to predict diverse human behaviour under weak supervision.

Research Experience (continued)

Feb 2020 – May 2024 **Ph.D., University of Melbourne** in Artificial Intelligence, supervised by A/Prof Nir Lipovetzky and Prof Charles Kemp. *AI and Autonomy Lab* and *Complex Human Data Hub*.

- Built the **symbolic foundations** of my current programme: **planning-based models** of human **problem solving** and **goal recognition**, embedding classical planners inside **Bayesian** models of intent and showing that action *timing* is a strong, previously overlooked signal for inferring goals.
- Validated the models against **controlled human experiments**, building the web-based experiment platform and analysing behavioural data in R and Python with **linear mixed-effects models**. Pairing formal planning machinery with behavioural evidence is what now lets me evaluate neurosymbolic systems against real cognition.

Mar 2013 – July 2015 **Research Assistant, Institute of Psychology, Chinese Academy of Sciences** in Neuroscience.

- Built a MATLAB platform that let participants **visualise their fMRI signals in real time**, implementing the **signal processing pipeline** behind the feedback loop; the origin of my empirical grounding in human behaviour.

Education

2020 – 2024 **Ph.D., University of Melbourne** in Artificial Intelligence.
Thesis title: *Planning and Goal Recognition in Humans and Machines*

2018 – 2019 **Master of Information Technology, University of Melbourne** in Computing.
High Distinction

2010 – 2014 **Bachelor of Science, Peking University** in Psychology.
Minor in Statistics

Selected Research Publications

Journal Articles

1. Z. Cai, C. R. Cardenas, K. Leo, **C. Zhang**, K. Backman, H. Li, B. Li, M. Ghorbanali, S. Datta, L. Qu, et al., "NEUSIS: A compositional neuro-symbolic framework for autonomous perception, reasoning, and planning in complex UAV search missions," *IEEE Robotics and Automation Letters*, vol. 10, no. 9, pp. 9502–9509, 2025. DOI: 10.1109/LRA.2025.3592098 ***Co-first author**.
2. Z. Li, **C.-y. Zhang**, J. Huang, Y. Wang, C. Yan, K. Li, Y.-w. Zeng, Z. Jin, E. F. Cheung, L. Su, et al., "Improving motivation through real-time fMRI-based self-regulation of the nucleus accumbens," *Neuropsychology*, vol. 32, no. 6, pp. 764–776, 2018. DOI: 10.1037/neu0000425
3. Y. Wang, Y. Deng, Z. Li, X. Li, **C.-y. Zhang**, Z. Jin, M.-x. Fan, M. T. Compton, E. F. Cheung, K. O. Lim, et al., "A trend toward smaller optical angles and medial-ocular distance in schizophrenia spectrum, but not in bipolar and major depressive disorders," *PsyCh Journal*, vol. 5, no. 4, pp. 228–237, 2016.
4. R.-t. Zhang, T.-x. Yang, Y. Wang, Y. Sui, J. Yao, **C.-y. Zhang**, E. F. Cheung, and R. C. Chan, "Structural neural correlates of multitasking: A voxel-based morphometry study," *PsyCh Journal*, vol. 5, no. 4, pp. 219–227, 2016. DOI: 10.1002/pchj.137

Conference Proceedings

1. S. Huang, **C. Zhang**, F. Ke, Z. Cai, G. Haffari, L. Qu, and H. Rezatofighi, "Mini-BEHAVIOR-Gran: Revealing U-shaped effects of instruction granularity on language-guided embodied agents," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics, 2026.
2. Q. Shen, G. Hu, and **C. Zhang**, "Proactive assistance agent with online goal recognition," in *Proceedings of the 36th International Conference on Automated Planning and Scheduling (ICAPS)*, vol. 36, 2026, pp. 275–283. DOI: 10.1609/icaps.v36i1.42837
3. **C. Zhang**, S. Huang, H. Rezatofighi, M. Vered, and B. Say, "Neurosymbolic active goal recognition in partially observable environments," in *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, Paphos, Cyprus: IFAAMAS, 2026, ISBN: 979-8-4007-2317-9.
4. **C. Zhang**, K. Ip, H. Rezatofighi, B. Say, and M. Vered, "A probabilistic framework for hierarchical goal recognition," in *Proceedings of the 23rd International Conference on Principles of Knowledge Representation and Reasoning (KR)*, IJCAI Organization, 2026, pp. 688–698. DOI: 10.24963/kr.2026/65
5. W. Li, **C. Zhang**, W. Li, G. Hu, and Y. Xu, "Modeling higher-order human beliefs using the justified perspective model," in *Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems*, ACM, 2025, pp. 1–8. DOI: 10.1145/3706599.3720223
6. **C. Zhang**, C. R. Cardenas, H. Rezatofighi, M. Vered, and B. Say, "Probabilistic active goal recognition," in *Proceedings of the 22nd International Conference on Principles of Knowledge Representation and Reasoning (KR)*, IJCAI Organization, 2025, pp. 880–890. DOI: 10.24963/kr.2025/85
7. **C. Zhang**, Y. Liu, D. Kulic, P. Carreno-Medrano, and M. Burke, "Modeling human sequential decision-making in the tower of london: Incorporating individual differences and timing-based replanning inference," in *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*, vol. 47, 2025.
8. **C. Zhang**, C. Kemp, and N. Lipovetzky, "Human goal recognition as bayesian inference: Investigating the impact of actions, timing, and goal solvability," in *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, IFAAMAS, 2024, pp. 2066–2074, **Pragnesh Jay Modi Best Paper Award**.
9. **C. Zhang**, C. Kemp, and N. Lipovetzky, "Goal recognition with timing information," in *Proceedings of the 33rd International Conference on Automated Planning and Scheduling (ICAPS)*, vol. 33, 2023, pp. 443–451. DOI: 10.1609/icaps.v33i1.27224
10. **C. Zhang**, N. Lipovetzky, and C. Kemp, "Comparing AI planning algorithms with humans on the tower of london task," in *Proceedings of the 45th Annual Meeting of the Cognitive Science Society*, vol. 45, 2023.

Manuscripts under Review

1. S. Huang, **C. Zhang**, F. Ke, Z. Cai, N. Rastgoo, G. Haffari, and H. Rezatofighi, *What we talk about when we talk about LLM planning: Evidence for two distinct planning abilities*, arXiv preprint arXiv:2607.11197, 2026. arXiv: 2607.11197.

Teaching, Supervision and Service

Research Funding and Fellowships

2026	**DAAD AINet Fellowship**, German Academic Exchange Service. Competitively awarded fellowship supporting an international AI research residency in Germany.
2024, 2026	Secured **Monash Summer Vacation Research Scholarship** to fund and supervise research students.

Teaching, Supervision and Service (continued)

2025 — Contributed to the successful continuation proposal for the **DARPA ANSR**-funded HARNESS programme.

Research Supervision

2025 - current — Co-supervise a **PhD candidate** at Monash University on neuro-symbolic planning and control.

2024 - current — Supervise Master's research projects at the University of Melbourne and Monash University on neuro-symbolic AI, goal recognition, multi-agent reinforcement learning, and symbolic belief modelling. Three supervised projects have reached peer-reviewed publication, at **KR** 2026, **ICAPS** 2026, and **CHI** 2025, with two further projects accepted at ICAPS 2026 workshops.

Selected Teaching Experience

2026 — **FIT5222 Planning and Automated Reasoning** (postgraduate). **Guest lecturer**, Monash University

- **COMP90054 AI Planning for Autonomy** (postgraduate). **Guest lecturer**, University of Melbourne
- **FIT5047 Fundamentals of Artificial Intelligence** (postgraduate). **Teaching Associate**, Monash University

2020-2022, 2024 — **COMP90054 AI Planning for Autonomy** (postgraduate). **Teaching Associate**, University of Melbourne

2020-2022 — **COMP90038 Algorithms and Complexity** (postgraduate). **Teaching Associate**, University of Melbourne

2020-2021, 2024 — **COMP30027 Machine Learning** (undergraduate). **Teaching Associate**, University of Melbourne

2021 — **COMP30024 Artificial Intelligence, COMP30026 Models of Computation** (undergraduate). **Teaching Associate**, University of Melbourne

Selected Awards and Achievements

2024 — **Pragnesh Jay Modi Best Paper Award**, International Conference on Autonomous Agents and Multiagent Systems

2023 — **Melbourne School of Engineering Travelling Scholarship**, University of Melbourne

- **Engineering and IT Conference Travel Scholarship**, University of Melbourne

2020 - 2023 — **Melbourne Research Scholarship**, University of Melbourne

2019 — **Grant of SummerTech LIVE**, The Victorian Government

2018, 2019 — **Dean's Honours List**, University of Melbourne

2012 — **Excellence Awards for Social Activity**, Peking University

Community Services

2026 — Co-organizer, **Human-Aware and Explainable Planning (HAXP)** and **Planning and Reasoning about Beliefs, Goals and Intentions (PR-BGI)** workshops, ICAPS 2026

2025 — Publicity Chair, ICAPS 2025

2024 — Organizer, Agents-Vic Autumn Symposium 2024

2022 - current — Program Committee member/Reviewer for top conferences and journals, including AAAI, AAMAS, ICAPS, CogSci, IJCAI, ICRA, KR, and Robotics and Autonomous Systems (RAS).